

Package ‘QuickDescriptives’

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Type Package

Title Quick Descriptive Summaries for Labelled Survey Data

Version 0.0.0.9000

Description A personal collection of helper functions for producing quick descriptive summaries -- crosstabs, missing-data reports, numeric summaries, and predictor-level breakdowns -- from labelled survey data imported via haven/labelled. Ported from the deprecated 'quackery' scratch project as a pilot for converting exploratory Quarto notebooks into a maintained package, documented separately in an accompanying Quarto site rather than in the function-defining notebooks themselves.

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purrr,
forcats,
stringr,
tibble,
rlang,
labelled,
huxtable,
openxlsx,
flextable,
stats,
utils,
haven

Suggests testthat (>= 3.0.0)

Config/testthat/edition 3

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count_across	<i>Frequency counts across several variables, stacked into one table</i>
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Description

Converts each of `vars` to a labelled factor and counts levels, stacking the results into a single long-format tibble (one row per variable-level combination) rather than returning a separate table per variable.

Usage

```
count_across(
  data,
  vars = NULL,
  empty_levels = FALSE,
  labels = TRUE,
  sort = FALSE
)
```

Arguments

<code>data</code>	A data frame containing the variables of interest.
<code>vars</code>	Character vector of variable names to summarise. If <code>NULL</code> (the default), every variable in <code>data</code> is used.
<code>empty_levels</code>	Logical. If <code>TRUE</code> , unused factor levels are kept in the output. Default <code>FALSE</code> .
<code>labels</code>	Logical. If <code>FALSE</code> , variable labels are stripped from the <code>var</code> column, leaving just the variable name. Default <code>TRUE</code> .
<code>sort</code>	Logical. If <code>TRUE</code> , sort each variable's levels by frequency. Default <code>FALSE</code> .

Value

A tibble with columns `var` (variable name, optionally with label), `levels`, `n`, and `p` (percentage).

ctab	<i>Two-way crosstab of one variable against a breakdown variable</i>
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Description

Two-way crosstab of one variable against a breakdown variable

Usage

```
ctab(
  data,
  vars = NULL,
  by,
  stat = "p",
  labels = TRUE,
  empty_levels = FALSE,
  drop_missing = "none"
)
```

Arguments

data	A data frame.
vars	Unquoted outcome variable to tabulate (data-masked, as in <code>dplyr::count()</code>).
by	Unquoted breakdown/crossbreak variable (data-masked).
stat	Which statistic to show in the crossbreak columns: "p" (percentage, default) or "n" (unweighted count).
labels	Logical. If TRUE (default), rename the first column to vars's variable label.
empty_levels	Logical. If TRUE, keep unused factor levels. Default FALSE.
drop_missing	Which missing data to drop before tabulating: "none" (default), "b" (both vars and by), "p" (predictor/by only), or "o" (outcome/vars only).

Value

A tibble with one row per level of vars, one column per level of by, plus a Total column.

ctabs	<i>Crosstab one or more outcomes against one or more predictors</i>
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Description

For each combination of vars (outcomes) and preds (predictors), builds a percentage crosstab with a base row and a total column showing the unconditional distribution of the outcome.

Usage

```
ctabs(data, vars, preds)
```

Arguments

data	A data frame.
vars	Character vector of outcome variable names.
preds	Character vector of predictor/crossbreak variable names.

Value

A nested list: one element per vars entry, each containing one tibble per preds entry. Each tibble carries a "tags" attribute (see `.add_tags()`) recording the predictor's variable label and the number of predictor-level columns, used by `ctabs_xlsx()` for layout.

ctabs_xlsx	<i>Export ctabs() output to a formatted Excel workbook</i>
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Description

Writes one worksheet per outcome variable, stacking each predictor's crosstab below the previous one with a merged predictor-label header row.

Usage

```
ctabs_xlsx(input, filename)
```

Arguments

input	The nested list returned by <code>ctabs()</code> .
filename	Path to the .xlsx file to write.

Value

Invisibly, the path written to (via `openxlsx::saveWorkbook()`).

df_miss	<i>Summarise missing data for every variable in a data frame</i>
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Description

Summarise missing data for every variable in a data frame

Usage

```
df_miss(data, more = FALSE)
```

Arguments

data	A data frame.
more	Logical. If TRUE, also break missingness down into system-missing vs. user-defined-missing (labelled "user NA") counts, for labelled variables. Default FALSE.

Value

A tibble with one row per variable: `n_miss`, `n_val`, `p_miss`, and (if `more = TRUE`) the same split into `na_*/user_*` columns.

<code>df_miss_brk</code>	<i>Missing data by variable, broken down by a grouping variable</i>
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Description

Reports missingness (or non-missingness) for every variable in data, crosstabulated by the levels of `brk`, plus a flag for whether each variable has any valid data at all within each level of `brk` and how many levels of `brk` it has valid data in.

Usage

```
df_miss_brk(
  data,
  brk,
  show = "perc_miss",
  keep_all_miss = FALSE,
  sort_miss = TRUE,
  form = "df",
  export = NULL
)
```

Arguments

<code>data</code>	A data frame.
<code>brk</code>	Character: name of the grouping/breakdown variable.
<code>show</code>	One of "perc_miss" (default), "n_miss", "n_not_miss", "p_not_miss".
<code>keep_all_miss</code>	Logical. If FALSE (default), drop variables with no valid data in any level of <code>brk</code> .
<code>sort_miss</code>	Logical. If TRUE (default), sort rows by how many levels of <code>brk</code> have valid data (descending).
<code>form</code>	Unused placeholder carried over from the original implementation (kept for interface compatibility).
<code>export</code>	Unused placeholder carried over from the original implementation (kept for interface compatibility).

Value

A tibble tagged (see [.add_tags\(\)](#)) with the `show` statistic and `brk`'s variable label, consumed by [df_miss_brk_h\(\)](#).

df_miss_brk_h	<i>Format df_miss_brk() output as a huxtable/flextable, with optional Excel export</i>
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Description

Format df_miss_brk() output as a huxtable/flextable, with optional Excel export

Usage

```
df_miss_brk_h(data, print = "f", export = NULL)
```

Arguments

data	Output of <code>df_miss_brk()</code> .
print	"f" (default) to return a flextable, "h" to return the underlying huxtable object.
export	If not NULL, a file path to also write a formatted .xlsx copy to.

Value

A flextable or huxtable object (see print).

mfreq	<i>Multi-variable frequency table (row or column percentages)</i>
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Description

Tabulates several variables that share a common response scale (e.g. a battery of trust/agreement items), returning either one row per variable (long = TRUE) or one column per variable (long = FALSE).

Usage

```
mfreq(data, vars = NULL, labels = FALSE, long = TRUE, stat = "p")
```

Arguments

data	A data frame.
vars	Character vector of variable names. If NULL (default), uses every variable in data (only sensible if they share response levels).
labels	Logical. If TRUE, use variable labels instead of names. Default FALSE.
long	Logical. If TRUE (default), one row per variable with levels as columns. If FALSE, one column per variable with levels as rows.
stat	Which statistic to show: "p" (percentage, default) or "n" (count).

Value

A tibble, shape depending on long, with a Base row/column giving the number of non-missing cases per variable.

natural_breaks*Cut several numeric variables into a shared set of breaks*

Description

Uses `cut()` to find a natural set of break points for `vars`, based on the first variable, then widens the outer breaks so no value in any of `vars` falls outside the range – or applies a user-supplied set of breaks (`brk`) directly.

Usage

```
natural_breaks(  
  data,  
  vars,  
  brk = NULL,  
  nbrk = 10,  
  stat = "p",  
  p_na = FALSE,  
  lab_style = 1  
)
```

Arguments

<code>data</code>	A data frame.
<code>vars</code>	Character vector of numeric variable names to bin. All share the same break points, so this works best when the variables have similar ranges/scales.
<code>brk</code>	Optional numeric vector of user-defined break points. If supplied, <code>nbrk</code> is ignored and these breaks are used for every variable in <code>vars</code> .
<code>nbrk</code>	Number of breaks to generate automatically. Default 10.
<code>stat</code>	Which statistic(s) to return: "p" (percentage, default), "n" (count), or "b" (both).
<code>p_na</code>	Logical. If FALSE (default), percentages are recalculated to exclude missing values. If TRUE, missing values are included in the percentage base.
<code>lab_style</code>	Integer 1 (default), 2, or 3, controlling how bin labels are formatted (e.g. "Greater than 0 and up to 10" vs. " $0 < \& \leq 10$ " vs. " $0-10$ ").

Value

A tibble with one row per bin and one column per variable (or per variable/statistic combination if `stat = "b"`).

num_crs	<i>Numeric summary statistics, broken down by a grouping variable</i>
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Description

Numeric summary statistics, broken down by a grouping variable

Usage

```
num_crs(data, nums, brk, stats = "all", order = "stat")
```

Arguments

data	A data frame.
nums	Character vector of numeric variable names to summarise.
brk	Character: name of the grouping/breakdown variable.
stats	Character vector of statistics to keep: any of "mean", "median", "sd", "min", "max", "n_missing", "n_valid". Default "all" keeps all of them.
order	"stat" (default) to sort by variable order as given, or "var" to sort alphabetically by variable then stat.

Value

A tibble with one row per variable/statistic combination, one column per level of brk, plus a Total column.

num_sum	<i>Numeric summary statistics for one or more variables</i>
---------	---

Description

Numeric summary statistics for one or more variables

Usage

```
num_sum(data, vars = NULL, wide = TRUE, user_na = TRUE, label = FALSE)
```

Arguments

data	A data frame.
vars	Character vector of numeric variable names. If NULL (default), uses every variable in data.
wide	Logical. If TRUE (default), one column per variable. If FALSE, one column per statistic.
user_na	Logical. If TRUE (default) and wide = TRUE, append a row tabulating any labelled user-defined missing values found in vars, which are otherwise excluded from the numeric statistics.
label	Logical. If TRUE, use variable labels instead of names. Default FALSE.

Value

A tibble, shape depending on wide.

return_one2	<i>One outcome summarised by one or more predictors, several statistics at once</i>
-------------	---

Description

Cross-tabulates a single outcome variable (out) by one or more predictors (preds), stacking together any combination of: number of valid cases (lev = "n_val"), percentage valid ("p_val"), percentage missing ("p_na"), percentage of cases matching a recoded set of levels (recode), and/or the mean of out (mean = TRUE).

Usage

```
return_one2(
  data,
  out,
  lev = NULL,
  recode = NULL,
  mean = FALSE,
  preds,
  pred_cols = NULL,
  user_na = TRUE,
  filter_na = "b"
)
```

Arguments

data	A data frame.
out	Character: name of the outcome variable.
lev	Character vector, any of "n_val", "p_val", "p_na", to include those missingness statistics as extra rows.
recode	Character vector of levels of out to collapse into a binary "in this group" indicator, reported as a percentage.
mean	Logical. If TRUE, also report the numeric mean of out (after stripping labels). Default FALSE.
preds	Character vector of predictor variable names.
pred_cols	Which of preds to place in columns rather than rows: NULL (default, uses preds[1]), a numeric index into preds, or a character vector of predictor names.
user_na	Logical. If FALSE, labelled user-defined missing values in preds are kept as valid categories rather than converted to NA. Default TRUE.
filter_na	Which missing data to drop before summarising: "b" (both out and preds, default), "o" (outcome only), "p" (predictors only), or "n" (don't filter).

Value

A named list with "Outcome label", "Predictors in columns", "Table" (the summary tibble), and "Names of preds in rows", in the form expected by [return_one2_format\(\)](#).

return_one2_format	<i>Format return_one2() output as a huxtable/flextable</i>
--------------------	--

Description

Format return_one2() output as a huxtable/flextable

Usage

```
return_one2_format(input, print = "h", colour = FALSE)
```

Arguments

input	Output of return_one2() .
print	"h" (default) to return a huxtable object, "f" to return a flextable.
colour	Logical. If TRUE, colour-scale the data cells (excluding the caption and header rows) by value. Default FALSE.

Value

A huxtable or flextable object (see print).

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